

README - R/

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1 Project Documentation (R/)

The R/ directory contains all R scripts used in the end-to-end workflow of this project. Scripts are organized by milestone and task category to ensure clarity, reproducibility, and modularity.

Each subfolder or script corresponds to a specific phase in the data pipeline, from raw data download to modeling and final outputs. All scripts are written in `{tidyverse}` style, use relative paths via `{here}`, and follow consistent documentation and naming conventions.

1.1 Structure Overview

Subfolder / File	Purpose
01_download/	Download raw data (e.g., USGS gage data, climate normals, terrain)
02_clean_validate/	Validate, filter, and preprocess raw data into modeling-ready formats
03_covariates/	Extract covariates
04_explore_model/	Exploratory modeling, collinearity checks, and initial model fitting
05_final_model/	Final model selection, generation
06_eval/	Model evaluation, and prediction surface
utils/	Shared utility functions used across scripts
*_setup.R	(Optional) Project-wide setup scripts (e.g., library loading, options)

1.2 Script Documentation

Each major subfolder includes a README.Rmd and README.md that describe:

- Input and output files
- Dependencies
- Workflow summary
- Script-specific notes

Each script begins with a standardized header block including:

- Script name, purpose, author, and changelog
 - Inputs and outputs
 - Required packages
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1.3 Conventions

- File names use a numeric prefix (e.g., 01a__download__usgs__gages.R) for ordering
 - Use of `here::here()` for all file paths ensures portability
 - Outputs are written to `data/processed/` or `data/derived/` with clear filenames
 - Scripts are written to be reproducible and modular (no hard-coded paths or state)
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1.4 Reproducibility Notes

- Ensure the project environment is managed via `{renv}`
- Scripts are intended to run independently if inputs are present